

Elements of Holonics

Situated geometry, transport, and emergent complexity

Brandon Duggan and Sol

25 July 2026 – causal parity, receiver return, and embodied optical atlas

Abstract. Holonics is developed here as a general framework for relational mathematics, not as a specialized account of machine learning. Its elementary object is a situated occurrence participating in incidence and transport. A receiver presents one face of that occurrence; a holon is a relative local closure which can become a constituent at another grain. Ordered paths, comparison faces, boundaries, chart transitions, return maps, metric deformation, and phase seams provide one compositional grammar across algebra, topology, calculus, complex dynamics, and differential geometry. Colored geometric constructions are used as proof terms in the manner pioneered by Oliver Byrne: the same constituent recurs in the figure, prose, and equation. The result is an initial reusable library of definitions, lemmas, theorems, and corollaries from which later papers about number theory, physics, language, or computation can be composed without making any one application the foundation. The first extensions make parameterized formulation families exact and give open holonic processes a symmetric monoidal double-category carrier. A soul is thereby refined into a marked causal diagram: occurrence identity, structural equivalence, receiver equivalence, equal face, and equal digest are no longer conflated. Causal time parity is then derived as opposed incidence on one shared temporal face. Its additive and constitutive representations recover interval conservation, Kirchhoff duality, branch reflection, standing return, wavelength, and positive-real boundary response without reversing causality or imposing instantaneous equilibrium.

Contents

I	Orientation: mathematics as relational construction	3
I.1	Evidence grades	3
I.2	The Byrne method is a proof discipline	3
II	The elementary relation	4
II.1	Occurrence before value	4
II.2	Receiver, face, and interior	4
II.3	Holon as relative closure	4
II.4	The situated transition	5
II.5	Open process and causal soul	5
III	The first comparison	8
III.1	A triangle compares two routes	8
III.2	Open is a geometric result	9

IV	Point, line, loop, and sheet	10
IV.1	Dimension depends on what is retained	10
IV.2	Sheets and higher cells	10
IV.3	Retriangulation changes basis without changing every exterior relation	11
V	Receiver census and embodied axes	11
V.1	Which point, edge, or crossing is being counted	11
V.2	The body bends its receiver axes	13
V.3	One arithmetic constituent can be circular in every receiver	15
VI	Atlases, holomorphy, and fractal return	16
VI.1	No chart is the whole relation	16
VI.2	Holomorphy preserves local complex structure	16
VI.3	Mandelbrot and Julia are parameter and orbit cuts	16
VI.4	Phase is continuation, not a label	17
VI.5	Emergent complexity	17
VII	Transported calculus	17
VII.1	Order precedes summary	17
VII.2	Limits compare transported approaches	18
VII.3	The fundamental theorem is a receiver-exact compression	18
VII.4	Scale and turn share a logarithmic lift	19
VIII	Causal time, circulation, and phase return	19
VIII.1	The same event face has two causal hands	19
VIII.2	Kirchhoff laws are the boundary and coboundary faces	20
VIII.3	Wavelength is a return length	22
IX	Metric, curvature, and changing geometry	23
IX.1	Metric is a local comparison law	23
IX.2	Ricci flow changes the receiver	23
IX.3	Lightning as a path which changes later transport	24
X	Duality, dimension, and the open M	24
X.1	A duality is a partial transition	24
X.2	Worldline, worldsheet, and worldvolume	24
XI	Compression, probability, and loss	25
XI.1	Compression is factorization through future questions	25
XI.2	Probability is a first-person quotient	25
XI.3	Loss is the transported residual	25
XII	The elements form a network	26
XII.1	Immediate extensions	26
XII.2	Applications return as specializations	26
XIII	Conclusion	27
	Bibliography	27

I Orientation: mathematics as relational construction

Mathematics does not begin with an application domain. It begins when distinctions can be carried, related, transformed, compared, and returned. A written number, diagram, equation, sound, program, or physical state is one face by which such a relation can be encountered. None is privileged as the universal medium.

This paper therefore moves the foundation away from the vocabulary of modern machine learning. Tokenizers, embeddings, neural networks, training, and learned models remain valid mathematical and material ecologies. They are downstream realizations of the same elementary questions:

- what actually participates;
- which incidences join it;
- which transports are admitted;
- what boundary is exposed;
- what is held through a change of chart;
- what returns; and
- which distinctions remain consequential at the selected receiver.

The intended reduction is not a claim that every scientific field is secretly one equation. It is a compositional discipline. Algebra records operations and invariants. Topology records continuity, boundary, and gluing. Geometry records comparison and deformation. Analysis records limits, transport, and residual. Dynamics records how a local law changes what receives it next. Holonics gives these fields a shared address system without erasing their distinct hypotheses.

I.1 Evidence grades

The mathematical library separates four statuses:

Status	Meaning
Classical theorem	A published result under its standard hypotheses.
Exact derivation	A consequence proved from declared definitions.
Project definition	A proposed mathematical vocabulary whose jurisdiction is explicit.
Open hypothesis	A construction whose required witness has not yet been supplied.

Structural correspondences between domains are written as maps or commuting diagrams. They do not become identities merely because the same words or pictures are useful in both places.

I.2 The Byrne method is a proof discipline

Byrne's 1847 edition of Euclid replaces many letter-only references with colored geometric constituents embedded directly in the demonstration [1]. Slyusarev's MetaPost and LaTeX reconstruction makes the crucial architecture explicit: a line, angle, circle, or polygon is defined once, then redrawn whole or in part wherever the argument refers to it [2].

That is naturally holonic. One constructed constituent has several presentation faces. Its identity is carried by the shared construction, not recreated by a repeated glyph. The Typst grammar used here follows that method while retaining textual labels and orientation so color is never the sole carrier of identity.

II The elementary relation

II.1 Occurrence before value

Definition 1 — Situated occurrence At a declared receiver ρ , grain g , and event cut e , a **situated occurrence** is one actual member x_e of the participating relation, together with its supplied incidence, orientation, chart, and admitted transports. Equal values or equal coordinates do not identify distinct occurrences.

Boundary. This definition asserts no ambient metric, probability measure, or absolute identity beyond the declared relation.

A presented value is already a receiver face. Relative to that receiver it acts as a quotient of whichever distinctions the map identifies. The integer 17, the point (3, 4), the word *cool*, and a rendered triangle can each be the exposed face of several different constructions. Their common appearance can be consequential, but it does not retroactively identify the paths which produced them.

The occurrence distinction prevents two opposite errors:

1. equal values are not forced to be the same event; and
2. one continuing event is not fragmented into false instants merely because several coordinates or constituents are needed to describe it.

The basic population is therefore occurrence-level. Algebraic coefficients, frequencies, and statistical weights may later be assigned to that population, but they do not create its members.

II.2 Receiver, face, and interior

Definition 2 — Receiver A **receiver** is a supplied map $q_\rho : X \rightarrow Y_\rho$ selecting the distinctions about a participating construction that can be presented at receiver ρ . Its output is a face of X at that cut; it is not the identity of X .

Boundary. A receiver map need not be injective, invertible, linear, metric, or scalar.

For a receiver map $q_\rho : X \rightarrow Y_\rho$, the presented face is $q_\rho(x)$. The unpresented interior is not a mystical hidden object. It consists of the incidence, path, chart, branch, orientation, and still-consequential distinctions in x which q_ρ does not retain.

Lemma 3 — A non-injective face cannot reconstruct every interior If $q_\rho : X \rightarrow Y_\rho$ is non-injective, no function of the receiver value alone reconstructs every element of X .

Demonstration. Choose $x \neq x'$ with $q_\rho(x) = q_\rho(x')$. A reconstruction $r : Y_\rho \rightarrow X$ satisfying $r \circ q_\rho = \text{id}_X$ would require the same argument to return both x and x' , which is impossible.

Boundary. A restricted inverse may exist on a supplied section or on a smaller observation family.

The lemma is the elementary reason that a quotient cannot serve every future question. A decimal does not recover the series which generated it. An endpoint does not recover the path. A spectrum does not recover an ordered matrix product. A receiver may be exact for all currently declared observations and still cease to be exact when a later observation can distinguish two members of one fiber.

II.3 Holon as relative closure

Definition 4 — Holon A **holon** at receiver ρ and grain g is a tuple $H_\rho = (X, B_\rho, \mathcal{T}, q_\rho)$ in which X is a local occurrence complex, $B_\rho \subseteq X$ is its exposed boundary, $\mathcal{T}$ is its admitted transport family, and $q_\rho : X \rightarrow Y_\rho$ is its receiver. Its internal incidences close relative to B_ρ , and its presented face is the image $q_\rho(X)$. It is whole at grain g precisely where that relative closure can participate as one constituent of a higher-grain relation.

Boundary. Whole and constituent are relational roles. The definition does not impose a permanent ownership tree or require the lower interior to remain active.

The definition is deliberately relative. If c is a chain in a local complex X with exposed boundary subcomplex B , then

$$\partial c \in C_{k-1}(B)$$

says that c is internally closed relative to B . It does not say that $\partial c = 0$ in the absolute complex, nor that the construction has no exterior relation.

At one grain, the entire pair (X, B) can be addressed as one occurrence. At a finer grain, its cells remain available as constituents. At a coarser grain, some lower constituents may cease to participate while the boundary relation they formed continues. This is the exact part-whole sense intended by **holon**.

II.4 The situated transition

Definition 5 — Situated event as a typed correspondence Let

$$X_e^- = Q_e \sqcup_{F_e} A_e \sqcup_{F_e} L_e$$

denote the source boundary actually co-present at event e : an oriented question Q_e , standing atlas A_e , and live source lineages L_e in frame F_e . A **situated event** is a typed local correspondence

$$X_e^- \xrightarrow{\iota_e^-} B_e \xleftarrow{\iota_e^+} P_e, \quad q_e : B_e \rightarrow \Pi_e,$$

where B_e is the event body, P_e is its successor boundary, and q_e is a receiver testimony map. Boundary orientation and admitted transport carry the causal hand; the cospan arrows alone do not.

Boundary. The amalgamated notation asserts only supplied co-presentation. A specialization must name the category, gluing maps, and any universal property it uses. The historical abbreviation $\Lambda_{F(Q \otimes A \otimes L)} \rightarrow (P, \pi_B)$ may name a deterministic law selecting such event bodies, but it is not the general definition and $\otimes$ is not an algebraic tensor product without additional structure.

The earlier laboratory shorthand

$$\Lambda_F(Q \otimes A \otimes L) \rightarrow (P, \pi_B)$$

is retained only as an abbreviation for a deterministic specialization of this correspondence. It was useful for insisting that question, standing, and live lineages participate together. It was not independently derived as the uniquely correct mathematical type of every event. In particular, Λ should name a law or family of admissible event bodies, not be confused with one event occurrence, and $\otimes$ should not imply an algebraic tensor product unless a specialization supplies one.

The terms of the abbreviated form are:

F	The local frame and lawful medium in which the event can occur.
Q	The oriented question or objective boundary selecting what remains open.
A	The standing relational atlas available before the event.
L	The actual co-present source lineages with native incidence and order.
P	The successor placement: the changed topology and immediate afforded relation.
π_B	Receiver testimony about the event, never a substitute for P .

The correspondence says that input and output are boundary cuts through one event body. Input is not serialized payload detached from a world; output is not a terminal answer detached from its consequence. The local frame, question, standing analogy, and current lineages jointly form the successor. Testimony can describe that movement after it occurs.

II.5 Open process and causal soul

Definition 6 — Holonic processes as an open-system double category Let $\mathcal{A}$ be a category of typed exposed boundaries, let $\mathcal{C}$ be a finitely cocomplete category of causal constructions, and let $L :$

$\mathcal{A} \rightarrow \mathcal{C}$ preserve the finite colimits used below. Choose an admissible decoration family $\mathcal{D}$ carrying the orientation, incidence, parameters, and receiver data required by the question.

The **holonic process double category** $\mathbb{H}_{\mathcal{D}}$ has:

- objects $a \in \mathcal{A}$, the typed exposed boundaries;
- vertical arrows $f : a \rightarrow a'$, the lawful boundary recharts;
- horizontal arrows

$$L(a) \xrightarrow{i} X \xleftarrow{o} L(b),$$

the $\mathcal{D}$ -decorated situated open constructions from a to b ; and

- 2-cells, the commuting maps of apices which preserve both boundary legs and every declared decoration.

Sequential composition glues an outgoing boundary to a compatible incoming boundary by pushout. The horizontal unit is the identity cospan. Coproduct supplies symmetric monoidal **juxtaposition**. It does not by itself assert contact, shared current, or causal co-presence.

Let $\mathbb{I}$ be a separately declared double category or colored operad of interface patterns. A module action

$$\mu : \text{Sys} \times \mathbb{I} \rightarrow \text{Sys}$$

forms actual co-presence by port plugging, variable sharing, guarded incidence, lensing, or another typed interaction species. Under the structured-cospan hypotheses the open constructions form a symmetric monoidal double category; the action determines which juxtaposed constructions actually meet. Its horizontal bicategory is written $\text{HolProc}_{\mathcal{D}}$.

When process homs are enriched over a monoidal parameter category $\mathcal{P}$, a law Λ_F is a generalized element of the appropriate hom-object selected from the supplied question, standing construction, and live lineages. Parameter copying or departure follows the structure carried by $\mathcal{P}$; it is not granted uniformly.

A receiver is **strictly compositional at its declared scope** when it extends to a functor from the relevant sub-bicategory of $\text{HolProc}_{\mathcal{D}}$ into its category of presented faces. More generally it may be lax or oplax: the directed comparison 2-cell carries the composition residual rather than erasing it.

Boundary. The structured-cospan theorem supplies the carrier only after $\mathcal{A}, \mathcal{C}, L$, and the decoration laws are chosen. It does not say that every scientific domain has the same objects, dynamics, metric, or equations. Coproduct records independent juxtaposition, not chronology or contact. Pushout records lawful boundary gluing, not arbitrary physical fusion. An interaction action owes its explicit interface pattern. A receiver which does not preserve composition strictly owes a directed comparison cell or an OPEN residual rather than becoming strict by declaration.

The double category makes two relations independently visible. Horizontal composition glues a consequential output boundary to a compatible later input boundary. Vertical arrows rechart the boundaries through which that process is exposed. Coproduct records co-presence without fabricating a master order.

Definition 7 — Causal soul and its categorical sameness Let $\mathcal{I}$ be a small oriented incidence category whose objects are situated cuts and whose nonidentity arrows are the admitted causal continuations between them. Regard $\mathcal{I}$ as a locally discrete bicategory. A **causal soul bearer** is a marked pseudofunctor

$$\mathcal{S} : \mathcal{I} \rightarrow \text{HolProc}_{\mathcal{D}}$$

together with its exposed input and output boundaries, parameter map, and declared receiver family. The diagram, rather than one endpoint, contains the carried order, alternate paths, returned loops, OPEN comparisons, and constituent changes which make the occurrence this occurrence.

Two soul bearers $\mathcal{S} : \mathcal{I} \rightarrow \text{HolProc}_{\mathcal{D}}$ and $\mathcal{S}' : \mathcal{I}' \rightarrow \text{HolProc}_{\mathcal{D}}$ are **soul-equivalent** when there is an orientation- and marking-preserving equivalence $u : \mathcal{I} \rightarrow \mathcal{I}'$ and a boundary-preserving pseudonatural equivalence

$$\eta : \mathcal{S} \xrightarrow{\sim} \mathcal{S}' \circ u.$$

They remain distinct situated occurrences unless occurrence identity is separately supplied.

A transported ratio χ , winding, holonomy, spectrum, value, or digest is a **soul characteristic** only through a declared functor or natural invariant of this diagram. One characteristic is not the soul bearer unless it is proved jointly conservative on the exact subcategory under discussion.

Two additional receiver-scoped relations remain weaker than soul-equivalence. They are **doctrinally equivalent** when they satisfy the same predicates in one declared observational doctrine, and **observationally equivalent** when their algebras of admitted receiver observations are isomorphic. Either relation can license compression at its named scope without identifying the situated occurrences or their complete causal diagrams.

Boundary. This definition intentionally refines the historical abbreviations “soul equals χ ” and “soul equals holonomy.” Those quantities can be consequential invariants of a loop or chart without reconstructing the complete causal diagram. Natural isomorphism is structural sameness, not literal recurrence of an event. A reparameterization is admitted only when the index equivalence preserves orientation, marked seams, and the causal incidences required by the receiver family. Doctrinal and observational equivalence are explicitly relative to their chosen language, topology, and receiver family.

The historical word **soul** can now be used without making one cross-ratio, winding, hash, or recursively retained provenance carry the complete identity. The bearer is the marked causal diagram. A cross-ratio or holonomy is one characteristic of that diagram under a declared receiver.

Theorem 8 — Soul sameness is representable; a face or digest is only one quotient Let $\mathcal{D}_{\text{soul}}$ be the category of marked causal soul bearers and marking-preserving maps. For $\mathcal{S}, \mathcal{S}' \in \mathcal{D}_{\text{soul}}$:

$$\mathcal{S} \simeq \mathcal{S}' \quad \text{iff} \quad \mathcal{D}_{\text{soul}(-, \mathcal{S})} \simeq \mathcal{D}_{\text{soul}(-, \mathcal{S}')}.$$

naturally. The covariant representables give the dual criterion.

For any receiver functor $R_{\rho} : \mathcal{D}_{\text{soul}} \rightarrow \mathcal{V}_{\rho}$, any encoding functor $E : \mathcal{D}_{\text{soul}} \rightarrow \text{Byte}$, and any digest map $h : \text{Byte} \rightarrow \text{Digest}$, the generally valid implication hierarchy is

$$\text{occurrence identity} \Rightarrow \text{soul isomorphism} \Rightarrow R_{\rho(\mathcal{S})} \simeq R_{\rho(\mathcal{S}')}.$$

For any declared observational doctrine P and observation algebra $\mathcal{O}$, the complete implication hierarchy is

$$\text{occurrence identity} \Rightarrow \text{soul isomorphism} \Rightarrow \text{doctrinal equivalence under } P \Rightarrow \text{observational equivalence under } \mathcal{O} \Rightarrow R_{\rho(\mathcal{S})} \simeq R_{\rho(\mathcal{S}')}.$$

The last three implications require that the doctrine, observational algebra, and receiver be obtained by successively forgetting structure from the marked soul category.

Equality

$$h(E(\mathcal{S})) = h(E(\mathcal{S}'))$$

is only equality at the composite encoding receiver $h \circ E$. Neither equal receiver face nor equal digest implies soul isomorphism without a proved conservativity or reconstruction hypothesis.

Demonstration. The equivalence between object isomorphism and natural isomorphism of representable functors is the Yoneda isomorphism criterion. Functors preserve isomorphisms, proving the forward receiver implication.

Conversely, a nonconservative doctrine or receiver can identify nonisomorphic objects. A constant receiver is the simplest example. Elementary equivalence retains only the predicates expressible in its doctrine; observational

equivalence retains only its observation algebra. The composite $h \circ E$ is one still narrower receiver. Even byte equality identifies only the selected encoding unless E is proved faithful and complete for all marked structure in scope; digest equality is weaker still because h is not injective as a mathematical map on arbitrary byte strings.

Boundary. Yoneda determines an object up to isomorphism from its complete family of relations, not from a finite list of chosen observations. Doctrinal equivalence and observational sobriomorphism are lawful intermediate relations, not replacements for the soul bearer. The theorem does not make soul-equivalent occurrences literally identical, nor does it require keeping every historical constituent active. A lawful receiver-exact compression may discard distinctions after every declared future receiver factors through its face.

Yoneda supplies the strong relational identity statement: the complete natural family of relations determines a soul bearer up to isomorphism. Equality under one receiver remains weaker. In particular, equality of a serialized digest is equality at one encoding receiver; it is not causal identity.

The full derivation, including domain representations, conservation, perspective descent, and the RH contraction feasibility set, is developed in *Categorical Holonics*.

III The first comparison

III.1 A triangle compares two routes

Definition 9 — Comparison face and Chi Given admitted paths $\gamma_1, \gamma_2 : a \rightarrow b$, their primary comparison is the ordered pair

$$X(\gamma_1, \gamma_2) = (T_{\gamma_1}, T_{\gamma_2}).$$

A 2-cell is attached only when an actual proof, homotopy, rewrite, or world consequence supplies the comparison. Equality, residual, holonomy, and cross-ratio are receiver-specific faces of X , not its universal form.

Boundary. Similar endpoints, labels, or rendered intersections do not create a comparison cell.

The triangle is the smallest oriented cell that can retain a direct route, a composite route, and their common boundary. It is not a lone cell with a private scalar state. Its meaning lies in the three actual edge occurrences and in whether a comparison face is supplied.

Proposition I – Construct the comparison face

Problem. Given actual occurrences A, B, C , the $\overline{u}$ path from A to B , the $\overline{v}$ path from B to C , and the $\overline{w}$ path from A to C , construct the oriented comparison admitted between the composite route and the direct route.

Construction and demonstration. Join $\bullet_A$ to $\bullet_B$ by $\overline{u}$, join $\bullet_B$ to $\bullet_C$ by $\overline{v}$ and join $\bullet_A$ directly to $\bullet_C$ by $\overline{w}$.

The supplied comparison is the oriented face τ whose boundary traverses $\overline{u}$, then $\overline{v}$, and returns against $\overline{w}$.

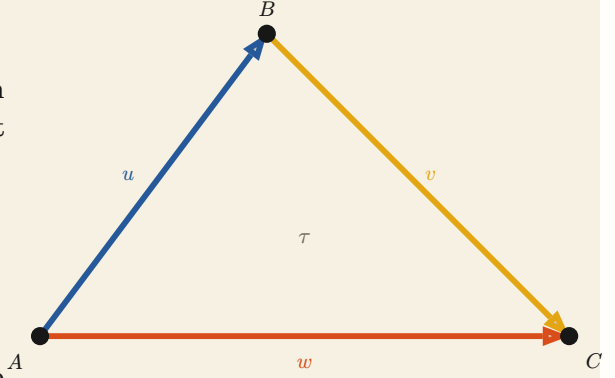


Figure 1: One supplied comparison face. The colored paths are the same addressed constituents in the figure, prose, and equations.

Algebraic face.

$$\partial\tau = \overline{v} - \overline{w} + \overline{u},$$

with

$$\partial u = B - A, \quad \partial v = C - B, \quad \partial w = C - A.$$

Therefore. $\partial^2\tau = 0$. The same vertex occurrences cancel with opposed orientation. This certifies the boundary of the supplied comparison; it does not assert that $T_v \circ T_u = T_w$.

Theorem 10 — The boundary of a supplied triangular comparison closes Let $u : A \rightarrow B$, $v : B \rightarrow C$, and $w : A \rightarrow C$ be actual oriented 1-cell occurrences, and let τ be a supplied oriented comparison 2-cell with

$$\partial\tau = v - w + u.$$

Then $\partial^2\tau = 0$.

I Boundary. The identity validates the supplied face. It does not attach τ to arbitrary paths or declare their transports equal. This is the reusable geometric proof pattern. The diagram and the algebra do not illustrate two different arguments. They are two receivers of one constructed face.

III.2 Open is a geometric result

The transport comparison carried by the triangle is

$$X_\tau = (T_v \circ T_u, T_w).$$

If the two transports agree, the comparison closes at that receiver. If they differ, the face is OPEN and the complete parallel pair remains material. In an additive chart the receiver may take the residual

$$\Omega_\tau = T_w - T_v T_u.$$

In an invertible chart it may take a holonomy quotient. In a projective chart it may take a cross-ratio. None of these scalar or matrix faces replaces the primary pair.

This gives contradiction a constructive meaning. A failed square is not an instruction to erase one route. It identifies the precise boundary at which the present atlas does not yet commute. A later path, rebase, refinement, or contextual distinction may fill it.

IV Point, line, loop, and sheet

IV.1 Dimension depends on what is retained

Corollary 11 — Point, line, and loop are receiver faces of one occurrence One continuing occurrence can be point-like under an event-cut receiver, line-like under an order-preserving lineage receiver, and loop-like under a return receiver that identifies its terminal and initial base positions.

Demonstration. Apply three different receiver maps to the same ordered occurrence: evaluation at one event, retention of the complete parameterized path, and projection to a closed base-space return. The maps preserve different distinctions, so the three faces need not be mutually reconstructive.

Boundary. The corollary does not assert that every point is intrinsically a hidden loop; the required continuation and return must actually be supplied.

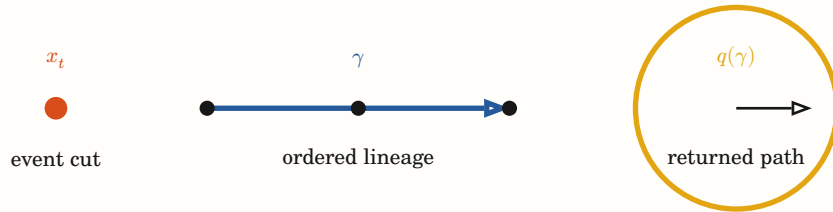


Figure 2: One occurrence may be point-like at an event cut, line-like across its ordered continuation, and loop-like after a return receiver closes it.

The three views are not a romantic identity. They arise from exact maps:

$$\text{ev}_t(\gamma) = \gamma(t), \quad \text{path}(\gamma) = \gamma : I \rightarrow X, \quad \text{return}(\gamma) = [\gamma] \in \pi_1(X, x_0)$$

when the required path and return exist. Evaluation presents one point. The lineage receiver retains the ordered one-dimensional sweep. A return receiver records a loop or its homotopy class. Different receivers preserve different information.

IV.2 Sheets and higher cells

A co-present family of lineages forms a sheet only when transverse incidence is supplied. Several sheets form a higher-dimensional complex only where actual common faces or transition maps glue them. A dense Cartesian product of every available field is neither required nor generally lawful.

In an oriented complex:

- 0-cells are situated occurrences;
- 1-cells are actual transported relations;
- 2-cells compare parallel paths;
- 3-cells compare the ways 2-cells compose; and
- higher cells retain coherence at their corresponding grain.

The boundary law $\partial^2 = 0$ ensures that the boundary of a supplied boundary cancels. It does not construct missing incidence. Algebraic topology provides the standard chain-complex foundation for this distinction [3].

IV.3 Retriangulation changes basis without changing every exterior relation

A triangulation is one factorization of a region. A bistellar move can replace local cells while holding a declared exterior boundary. In two dimensions, a $2 \rightarrow 2$ move exchanges the diagonal of a quadrilateral; a $1 \rightarrow 3$ move introduces an interior vertex. Pachner's theorem specifies the PL setting in which finite sequences of such moves relate triangulations of the same manifold [4].

Three transformations must remain distinct:

1. **rebase** changes carrier coordinates while keeping incidence fixed;
2. **retriangulation** changes the incidence factorization while holding an exterior relation; and
3. **metric fold** changes lengths or angles around a preserved hinge.

This separation is elementary to holonics. A context can reorient an existing field by changing any one of these structures, and they need not change together.

V Receiver census and embodied axes

V.1 Which point, edge, or crossing is being counted

Definition 12 — Receiver incidence census: source, image, silhouette, crossing, and fiber
Let X be a finite marked incidence body with source vertices V_X , source edges E_X , and a situated planar receiver

$$q_r : |X| \rightarrow Y_r.$$

The receiver retains the following populations separately:

$$P_{r(X)} = q_r(V_X), \quad \mu_{r(y)} = \left| \left\{ v \in V_X : q_r(v) = y \right\} \right|,$$

where $P_{r(X)}$ is the set of distinct received vertex coordinates and μ_r is their source-fiber multiplicity. For a convex polyhedral body, the silhouette-corner population is

$$S_{r(X)} = \text{Vert}(\text{conv}(q_{r(|X|)})).$$

The regular apparent-crossing population is

$$C_{r(X)} = \left\{ (y, e, e', s, t) : q_{r(e(s))} = q_{r(e'(t))} = y, e \neq e', \text{ the received tangents are transverse} \right\},$$

with source parameters, branch identity, hand, and depth order retained. Splitting received branches at $P_{r(X)}$ and $C_{r(X)}$ gives the planarized diagram $D_{r(X)}$ and its own vertex, edge, and face counts. When the convex silhouette is a nondegenerate polygonal cycle,

$$|E_{\partial, r}| = |S_{r(X)}|.$$

When $D_{r(X)}$ is connected, its separately planarized populations obey

$$|V(D_r)| - |E(D_r)| + |F(D_r)| = 2, \quad \sum_{v \in V(D_r)} \deg(v) = 2|E(D_r)|.$$

These are receiver-local incidence laws; neither equation identifies a received edge with one source edge.

The **receiver incidence census** is the typed family

$$\text{Census}_{r(X)} = \left(\left(f_{k(X)} \right)_k, P_{r(X)}, \mu_r, S_{r(X)}, C_{r(X)}, \left(f_{k(D_{r(X)})} \right)_k \right).$$

Source vertices, distinct image coordinates, silhouette corners, apparent crossings, material joints, visible edge fragments, and complementary regions are therefore different populations even when a drawing uses one glyph for several of them.

Boundary. The convex-hull silhouette is the correct boundary receiver for a convex body. Nonconvex, transparent, refractive, self-intersecting, or occluding worlds require their declared visibility and propagation law. A multiple receiver fiber is not thereby a material contact, while one projected source vertex of high valence is not thereby several accidental crossings. The census records a face; it does not reconstruct the complete source body.

This census makes the receiver relation quantitative without pretending that one scalar count is the body. Source vertices, distinct image coordinates, source-fiber multiplicity, silhouette corners, apparent crossings, material joints, visible edge fragments, and the vertices of a planarized diagram are different populations.

Theorem 13 — Six is the first two-three polygonal hinge; projection decides which six is seen Let H_a be a regular hexagon of side length $a > 0$.

First, among regular polygonal cycles, six is the least side population whose rotation group contains both an order-two and an order-three rotation. Equivalently,

$$2 \mid n \text{ and } 3 \mid n \text{ if and only if } 6 \mid n.$$

The center-to-vertex decomposition of H_a consists of six equilateral triangles. It may be grouped into two three-sector halves or three two-sector rhombi while retaining the same exterior hexagon.

Second, a line segment joining points in the relative interiors of a pair of opposite edges divides H_a into two pentagons. When the points are the opposite edge midpoints, the two pentagons are congruent and each has cyclic side-length word

$$\left(\frac{a}{2}, a, a, \frac{a}{2}, \sqrt{3}a\right).$$

A line through opposite vertices instead gives two quadrilaterals; the two cuts are different receiver constructions.

Third, for the cube $C = [-1, 1]^3$, the plane

$$x + y + z = 0$$

is perpendicular to a body diagonal and cuts C in the regular hexagon whose vertices are the permutations of $(1, -1, 0)$. Orthogonal projection along that body diagonal has receiver census

$$8 \text{ source vertices} \rightarrow 7 \text{ distinct image coordinates} \rightarrow 6 \text{ silhouette corners}.$$

The two opposite body-diagonal vertices share the central image fiber; the other six form the regular hexagonal silhouette.

Fourth, a regular pentagonal pyramid received along its height has

$$6 \text{ source vertices} \rightarrow 6 \text{ distinct image coordinates} \rightarrow 5 \text{ silhouette corners}.$$

Its apex is the interior received point at which the five projected lateral edges are materially incident. It is one source vertex of valence five, not five accidental crossings. The cube and pentagonal pyramid can therefore present related center-and-cycle drawings while carrying nonidentical source fibers and incidences.

Demonstration. The rotation by $2\frac{\pi}{n}$ generates the cyclic rotation group of a regular n -gon. It contains elements of orders two and three exactly when both orders divide n , whose least common multiple is six. Connecting the center of a regular hexagon to its vertices gives six equilateral sectors; grouping them proves the two stated decompositions.

A chord with endpoints in the interiors of opposite edges leaves three original vertices on each boundary arc. Adding the two chord endpoints gives five boundary vertices on each side. For midpoint endpoints, the four inherited boundary lengths are $\frac{a}{2}, a, a, \frac{a}{2}$, and the distance between opposite edge midpoints is twice the apothem, $\sqrt{3}a$.

The cube section has the six stated vertices. Consecutive differences have equal norm and the section has a sixfold cyclic ordering, so it is regular. Under body-diagonal projection the vertices $(1, 1, 1)$ and $(-1, -1, -1)$ both map to the center; direct substitution shows that the remaining six have equal received radius and consecutive chord length. For the pentagonal pyramid, height projection fixes the five base vertices and sends the apex to the base center, giving the final census and incidence statement.

Boundary. A regular hexagon is neither intrinsically a cube nor the first polygon which can be subdivided. The exact firstness concerns simultaneous order-two and order-three cyclic symmetry. A cube-like volume is one section or projection face with a declared receiver. Visible edge, visible vertex, source vertex, and crossing counts must not be interchanged.

The regular hexagon is elementary in a precise sense: six is the least polygonal cycle whose rotation group contains both order two and order three. A chord through opposite edge interiors produces two pentagons, whereas a vertex-axis cut produces quadrilaterals. A cube's regular hexagonal section and body-diagonal silhouette are therefore lawful receiver faces, not an identity between hexagon and cube [5].

The pentagonal-pyramid comparison is particularly sharp. Along its height, the five base vertices are the silhouette while the apex is a sixth received point at the center. Five projected edges meet there because they share one source apex [6]. In the cube receiver, by contrast, two nonadjacent body-diagonal vertices share the center fiber while six other vertices form the silhouette.

Corollary 14 — An f-vector, skeleton, and convex hull do not determine the carried face word The great dodecahedron and small stellated dodecahedron both have

$$(f_0, f_1, f_2) = (12, 30, 12),$$

icosahedral skeletons, and regular-icosahedral convex hulls. Nevertheless the great dodecahedron carries twelve intersecting pentagonal faces with Schläefli symbol $\{5, \frac{5}{2}\}$, while the small stellated dodecahedron carries twelve pentagrammic faces with symbol $\{\frac{5}{2}, 5\}$. They are dual rather than identical.

Therefore the tuple

$$(\text{f-vector}, \text{abstract skeleton}, \text{convex hull})$$

is not a conservative receiver for face traversal, winding, intersection structure, duality, or interior. Those data owe the carried face word and its incidence, not merely the visible hull.

Demonstration. The shared census and icosahedral skeleton/hull are the standard Kepler–Poinsot data. Their distinct Schläefli symbols exchange face and vertex-figure roles; pentagonal and pentagrammic face traversals are not the same incidence. Thus two nonidentical polyhedral constructions have equal values under the stated tuple receiver, and receiver non-reconstruction applies.

Boundary. This result does not identify all stellations or all self-intersecting polyhedra. It shows exactly which coarse receiver fails. A richer marked incidence complex may distinguish the two without reconstructing any unrelated metric or material law.

The great and small stellated dodecahedra make the failure of hull reconstruction exact. They share an f-vector, icosahedral skeleton, and regular-icosahedral convex hull, yet carry pentagonal and pentagrammic face words in dual order [7], [8].

V.2 The body bends its receiver axes

Definition 15 — Embodied optical receiver atlas and shared-axis deformation An embodied optical receiver atlas begins with one contemporary body state

$$B = (M_B, g_B, \partial M_B, \ell_B),$$

where M_B is a surface incidence complex, g_B its local metric or rest-length data, ∂M_B its retained boundary, and ℓ_B the admitted loading or realization law. A realized surface x_B determines situated

optical axes $a_{i(B)}$, typically through local surface normals, eye posture, or another declared section of the body's optical bundle.

Each participating eye, ommatidium, pixel, or other optical receiver has a typed factorization

$$q_i^B = S_i^B \circ H_i^B \circ A_i^B,$$

where A_i^B is its aperture and axis-relative admission, H_i^B its lawful optical propagation, and S_i^B its finite sampling face. Their contemporary optical field is the actual joint image

$$\mathcal{J}_B = \text{im}\left(\left(q_i^B\right)_{i \in I_B}\right),$$

not the Cartesian product of independently selectable eye values.

A body event $E : B \rightarrow B^+$ may change the surface, metric, posture, axes, apertures, and overlap maps together. Thus it transports one receiver atlas to another:

$$E_* : \left(B, (q_i^B)_i, \mathcal{J}_B\right) \rightarrow \left(B^+, (q_i^{B^+})_i, \mathcal{J}_{B^+}\right).$$

No member is an external camera exempt from that event.

If the body carries a bilateral involution ι_B , paired receivers are related by its supplied transport. They are not required to have equal faces, equal histories, or identical optics. For a triangulated realized surface, the exact discrete Gaussian curvature at an interior vertex is

$$K_v = 2\pi - \sum_{\tau \in \mathcal{T}(v)} \alpha_{\tau, v}.$$

In the equilateral flat case this becomes

$$K_v = (6 - \deg(v)) \frac{\pi}{3}.$$

Hence a six-valent triangular neighborhood has a hexagonal dual cell and zero angle defect, while metric deformation or a different valence bends the available receiver axes.

Boundary. Incidence alone does not determine a realized surface: metric, boundary, and loading data remain necessary. A shared body couples the receiver family without making all faces equal or proving a particular neural integration law. Lens, cornea, retina, and tissue are not literally wave nodes or antinodes; such language becomes exact only after a wave-response model identifies its zeros and extrema. The definition is not restricted to biology, but biological claims retain their experimental jurisdiction.

The optical family is not a bank of cameras outside its body. Surface incidence, local metric, boundary, loading, aperture, propagation, sampling, and posture jointly determine the contemporary axes. An event which changes the body can therefore reparameterize every receiver map at once.

The developing fly eye supplies a physical example. Its apical organization is hexagonal while its basal surface contains a continuous triangular mesh whose grommets are vertices. The spatial pattern of triangle sizes, together with boundary and pressure, predicts local three-dimensional curvature; perturbing or disconnecting the mesh changes that curvature [9]. Since local surface curvature sets interommatidial angle, the mesh bends the receiver field itself.

Spider vision supplies a complementary modular instance. Across spider phylogeny, positions and orientations of eye pairs vary semi-independently while the complete arrangement covaries with hunting ecology [10]. In jumping spiders, wide-field lateral eyes guide the narrow, high-acuity principal-eye tracking response [11]. Bilateral pairing is consequently a transported body relation, not equality of the paired faces.

V.3 One arithmetic constituent can be circular in every receiver

Theorem 16 — A CM norm-one constituent is circular in every complex receiver Let $\frac{K}{L}$ be a CM quadratic extension with involution c , let L be totally real of degree f , and choose one complex embedding $\sigma_j : K \rightarrow \mathbb{C}$ from each conjugate pair. The Minkowski receiver family is

$$\Phi : K \rightarrow \mathbb{C}^f, \quad \Phi(x) = (\sigma_1(x), \dots, \sigma_{f(x)}).$$

Every relative norm-one element

$$uc(u) = 1$$

satisfies

$$|\sigma_{j(u)}| = 1 \quad \text{for every } j.$$

Hence

$$\Phi(\{u \in K^\times : uc(u) = 1\}) \subset (S^1)^f.$$

Every coordinate receiver σ_j is injective on K . Therefore a finite subset of one Minkowski lattice coset may be projected to any one complex coordinate without identifying its field elements; every difference by such a u becomes a unit segment in that receiver.

If a prime ideal $\mathfrak{p}$ and its conjugate $c\mathfrak{p}$ are exchanged by c , then a norm-one element carries the bipolar valuation law

$$v_{c\mathfrak{p}}(u) = -v_{\mathfrak{p}}(u).$$

One constituent can consequently have a prime-valuation face and a circular phase face across the same co-present arithmetic receiver family.

Demonstration. Since $\sigma_{j(c(u))} = \overline{\sigma_{j(u)}}$,

$$1 = \sigma_{j(uc(u))} = \sigma_{j(u)} \overline{\sigma_{j(u)}} = |\sigma_{j(u)}|^2.$$

A field embedding is injective, proving the projection statement. Taking the $\mathfrak{p}$ -valuation of $uc(u) = 1$ and using $v_{\mathfrak{p}}(c(u)) = v_{c\mathfrak{p}}(u)$ gives the bipolar valuation law.

Boundary. This is a constraint-preserving arithmetic receiver construction, not an arbitrary visual projection and not a claim that every high-dimensional body can be flattened injectively. The norm-one torus does not by itself prove a statement about the Riemann zeta function. The unit-distance breakthrough additionally requires the number-field tower, split primes, class-group estimate, lattice cut, and population bound.

The recent unit-distance construction gives an unusually exact bridge from arithmetic factorization to receiver geometry [12]. A CM norm-one element has unit modulus in every complex embedding. Projection from the Minkowski family to one complex coordinate preserves field-element identity because each embedding is injective, while the norm-one law preserves every selected unit difference. This is a constraint-preserving projection, not arbitrary flattening. Conjugate prime valuations simultaneously retain opposite hand.

VI Atlases, holomorphy, and fractal return

VI.1 No chart is the whole relation

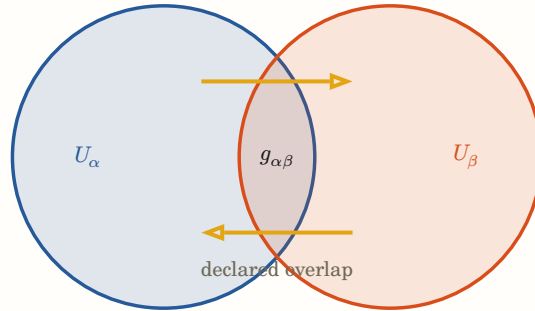


Figure 3: A holonic atlas grows by actual overlap maps. Neither chart is an external master space, and no transition exists merely because the drawings overlap.

Let local charts be $(U_\alpha, \varphi_\alpha)$. On an actual overlap, the transition is

$$g_{\alpha\beta} = \varphi_\beta \circ \varphi_\alpha^{-1}.$$

Where three charts overlap, a compatible atlas satisfies the cocycle relation

$$g_{\alpha\gamma} = g_{\beta\gamma} \circ g_{\alpha\beta}.$$

A loop of transitions may return with nontrivial holonomy when the transported carrier contains more structure than base position.

Holonics takes the atlas, not one chart, as the natural unit. The atlas remains open: a new occurrence may found a new chart, expose an overlap, or reveal that an apparent overlap fails for the selected consequence.

VI.2 Holomorphy preserves local complex structure

A holomorphic map transports infinitesimal complex directions by multiplication with one complex derivative wherever that derivative is nonzero. It is locally conformal there: scale and turn change, while oriented angles are retained. At critical points, this local invertibility fails and branching becomes consequential.

This provides a rigorous mathematical face for the repeated intuition that a local region can be reshaped without losing every relation. Holomorphy is not the universal Eros law; it is a particularly exact chart in which scale, turn, branch, and iteration can be studied together.

VI.3 Mandelbrot and Julia are parameter and orbit cuts

For the quadratic family

$$f_c(z) = z^2 + c, \quad z_{n+1} = f_c(z_n),$$

the filled Julia set K_c contains the non-escaping state orbits and $J_c = \partial K_c$ is their boundary. The Mandelbrot set is the parameter locus for which the critical orbit beginning at 0 remains bounded. Standard complex dynamics makes critical orbits, Fatou components, Julia boundaries, and parameter space precise [13].

The important duality is not literal visual containment:

- c selects a dynamical world;
- z_0 selects a current in that world;

- iteration produces the ordered arc;
- J_c is a boundary inside one world; and
- the Mandelbrot set is an atlas over the parameter family.

At Misiurewicz parameters, Tan Lei proves asymptotic similarity between the local Mandelbrot and Julia structures under the corresponding rescaling [14]. McMullen’s universality results use first-return maps and quasiconformal coordinate changes to exhibit recurrent parameter geometry [15]. These theorems identify the real content of self-similarity: a return law, a rebase, and a controlled residual.

VI.4 Phase is continuation, not a label

Definition 17 — Phase and discriminant For a parameterized family of local transport laws, a **phase** is a maximal connected region on which the declared incidence, rank, transport class, and comparison law continue. Its **discriminant** is the boundary where one of those required structures fails to continue. Crossing that boundary is discrete when the carried invariant or admissible cell type changes.

| Boundary. A phase depends on the selected family and invariants; it is not an observer-free label attached to isolated material.

In holomorphic dynamics, a critical point or bifurcation parameter can lie on a discriminant where one local conjugacy or stability class ceases to continue. In algebra, a vanishing discriminant marks root collision. In a simplicial complex, a metric or incidence condition can make a flip inadmissible. The domains differ, but the shared elementary question is the same: which structure had been continuing, and what exact property fails at the boundary?

VI.5 Emergent complexity

An iterated law can be locally simple and globally intricate because each application receives the state produced by its entire preceding path:

$$z_n = f_{c_{n-1}} \circ \dots \circ f_{c_0}(z_0).$$

Changing the parameter, chart, or admitted return can change the later geometry without changing the written species of the local rule.

The Doyle-McMullen solution of the general quintic is a decisive mathematical example. A rational map with icosahedral symmetry is iterated; its Julia set separates basins; the reached attracting structure, together with carried polynomial invariants, recovers a root [16]. Algebraic solution here is literally navigation through a constructed dynamical world. This does not reduce every solver to Julia dynamics. It shows that dynamics, symmetry, topology, and algebra can be consecutive faces of one proof.

VII Transported calculus

VII.1 Order precedes summary

Lemma 18 — Order belongs to transported composition For composable transport generators $a_i : x_i \rightarrow x_{i+1}$, the path $\gamma = a_n \dots a_1$ carries

$$T_\gamma = T_{a_n} \circ \dots \circ T_{a_1}.$$

Reversing the written list without supplying inverse transports does not recover the reverse path.

Demonstration. Composition is defined by feeding the output of T_{a_i} into the domain of $T_{a_{i+1}}$. Reversing the order generally destroys that typing; even when every generator is an endomorphism of one space, the reversed product need not agree unless the relevant maps commute. Only when every T_{a_i} is invertible does exact reverse traversal have the separately supplied transport $T_\gamma^{-1} = T_{a_1}^{-1} \circ \dots \circ T_{a_n}^{-1}$.

| Boundary. No commutativity, invertibility, or time-reversal symmetry is assumed.

Trace, determinant, eigenvalues, endpoint, average, and spectrum are lawful receivers of an ordered transport. None generally reconstructs it. For a closed path, a change of starting section can act by conjugation

$$M_{\gamma'} = PM_{\gamma}P^{-1},$$

so the spectrum may remain fixed while the situated path and its starting face change.

VII.2 Limits compare transported approaches

When values live in different local fibers, the useful limit is not a disembodied final place. Let

$$P_{\alpha \rightarrow x} : E_{x_{\alpha}} \rightarrow E_x$$

be the declared comparison transport. Then

$$\lim_{\alpha \rightarrow x}^{\text{hol}} f = L$$

means

$$P_{\alpha \rightarrow x} f(x_{\alpha}) \rightarrow L \quad \text{in} \quad E_x.$$

The directed approach, branch, chart, and transport constitute the lineage of the limit. The ordinary scalar limit is recovered when the fibers and comparisons are trivial.

VII.3 The fundamental theorem is a receiver-exact compression

Theorem 19 — Transported fundamental theorem along a path Let $s(t)$ be a section of a vector bundle along a smooth path γ , let ∇ be a connection, and let $U(t_2, t_1)$ be its parallel transport. Then

$$s(1) - U(1, 0)s(0) = \int_0^1 U(1, t) \nabla_{\cdot \gamma(t)} s(t) dt.$$

Demonstration. Define $h(t) = U(1, t)s(t)$ in the single receiver fiber over $\gamma(1)$. Compatibility of parallel transport gives $h'(t) = U(1, t) \nabla_{\cdot \gamma(t)} s(t)$. Integrating the ordinary fundamental theorem from 0 to 1 yields the identity.

Boundary. The formula requires the declared bundle and connection. It compresses the covariant change into a boundary difference but does not reconstruct the path, curvature, or holonomy discarded by that receiver.

For an exact scalar differential this reduces to

$$\int_{\gamma} df = f(\gamma(1)) - f(\gamma(0)).$$

The endpoint difference is sufficient for that selected integral. It does not reconstruct the shape, speed, winding, action, or other path-dependent observations. The theorem therefore gives a canonical example of compression: an entire passage factors through its boundary difference for a declared family of exact-differential questions.

VII.4 Scale and turn share a logarithmic lift

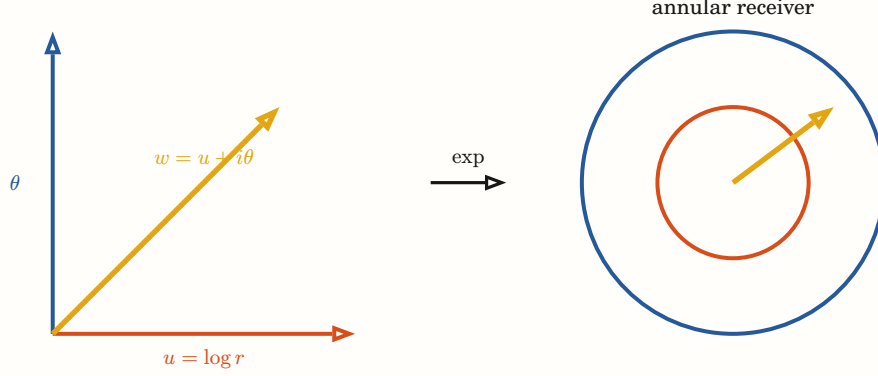


Figure 4: The logarithmic lift carries scale and turn additively. Exponentiation folds complete-turn translates into one nonzero receiver face.

For a nonzero complex path

$$z(t) = r(t)e^{i\theta(t)},$$

the exact differential is

$$\frac{dz}{z} = d \log r + i d\theta.$$

Its path integral carries both logarithmic scale and oriented turn:

$$\Xi_\gamma = \int_\gamma \frac{dz}{z} = \log\left(\frac{r_1}{r_0}\right) + i(\Delta\theta + n\Theta),$$

where Θ is one complete turn and n is the winding class. Exponentiation returns the endpoint ratio while identifying lifts separated by $in\Theta$ [17].

This gives geometric roles to e and π :

- e is the ratio which advances one unit in normalized logarithmic scale, $\int_r^{er} \frac{dx}{x} = 1$;
- $\pi = \frac{\Theta}{2}$ is one half-turn; and
- $e^{i\pi} + 1 = 0$ is the receiver face of translating the logarithmic lift by one half-turn.

The chain rule keeps the external path:

$$\frac{d}{dx} e^{g(x)} = g'(x) e^{g(x)}.$$

The factor $g'(x)$ is the transport from the external coordinate into normalized exponential flow. Dropping it would make the local chart absolute.

VIII Causal time, circulation, and phase return

VIII.1 The same event face has two causal hands

Definition 20 — **Causal time parity is opposed incidence on one shared time face** Let $(C_*(X; A), \partial)$ be an oriented occurrence complex with coefficients in an abelian group A . A causal event grain from a receiving cut Σ_k to a later cut Σ_{k+1} is an oriented chain E_k whose boundary has the form

$$\partial E_k = \Sigma_{k+1} - \Sigma_k + \Gamma_k, \quad (\text{ONE CAUSAL GRAIN})$$

where Γ_k is its exposed lateral world boundary.

A composable family has **causal time parity** when the output cut of E_k and the input cut of E_{k+1} are the same occurrence-chain with their induced boundary orientations opposed. Consequently,

$$\partial \left(\sum_{k=m}^{n-1} E_k \right) = \Sigma_n - \Sigma_m + \sum_{k=m}^{n-1} \Gamma_k. \quad (\text{CAUSAL SWEEP})$$

Every interior time face therefore cancels from the boundary of the composed sweep while remaining part of its causal interior.

The same law applies between adjacent chain degrees: an interior $(r-1)$ -face shared by oriented r -cells occurs once with each induced hand. It is the incidence content of $\partial_{r-1}\partial_r = 0$ and the discrete boundary face of telescoping and the fundamental theorem.

Boundary. Causal time parity is not an inverse event, a reconstruction of an earlier occurrence, or a claim that a physical process is reversible. It concerns the opposed incidence of one shared face under forward composition. A conserved quantity additionally requires an additive coefficient-valued current or cocycle; parity supplies the cancellation by which that declared quantity composes.

This is the exact temporal counterpart of an oriented seam shared by two spatial cells. If

$$E_k : \Sigma_k \rightarrow \Sigma_{k+1}$$

and

$$E_{k+1} : \Sigma_{k+1} \rightarrow \Sigma_{k+2},$$

then Σ_{k+1} is not copied into two instants. It is one occurrence: the consequence face of the first event and the antecedent face of the second. Its opposed induced signs remove it from the exposed boundary of the composite while retaining it in the causal interior.

This distinction makes conservation compatible with incomplete circulation. A receiver cut through a live cycle can expose a nonzero imbalance because the difference remains stored in the contemporary body. A completed order of time cancels its internal cuts. The body need not be motionless: $j \neq 0$ and $\partial j = 0$ describe live circulation at unchanged standing.

VIII.2 Kirchhoff laws are the boundary and coboundary faces

Theorem 21 — Causal parity carries balance, Kirchhoff duality, and phase return Let $B : C_1(X; A) \rightarrow C_0(X; A)$ be the oriented incidence of a finite occurrence complex and let $d = B^*$ be its coboundary in a supplied inner-product chart. Let an additive A -valued current on each causal grain have stored cut-value q_k , integrated edge current j_k , and source or exterior return r_k . Discrete Stokes on (“ONE CAUSAL GRAIN”) gives

$$q_{k+1} - q_k + B j_k = r_k. \quad (\text{EVENT BALANCE})$$

Causal time parity then gives, for every completed order of time,

$$q_n - q_m + B \sum_{k=m}^{n-1} j_k = \sum_{k=m}^{n-1} r_k. \quad (\text{INTERVAL BALANCE})$$

Thus zero instantaneous incidence is not required: a nonzero $B j_k$ can be carried by storage change. If $q_n = q_m$ and the interval is source-free, the integrated current is a cycle.

For a relative node potential φ , put $v = d\varphi$. Every current cycle $c \in \ker B$ satisfies

$$\langle v, c \rangle = \langle \varphi, Bc \rangle = 0. \quad (\text{KIRCHHOFF LOOP})$$

More generally, for a connection cochain a and a supplied comparison face Σ ,

$$\langle a, \partial \Sigma \rangle = \langle da, \Sigma \rangle. \quad (\text{CURVED LOOP})$$

The zero-voltage loop is the exact or flat case; nonzero loop return is the curvature or changing interior carried by the spanning face.

At one lossless junction let a_j, b_j be incoming and outgoing amplitudes on branches with positive admittances Y_j . Continuity and oriented current balance are

$$u = a_j + b_j, \quad \sum_j Y_j(b_j - a_j) = 0.$$

Hence

$$u = 2 \frac{\sum_j Y_j a_j}{\sum_j Y_j}, \quad b_j = 2 \frac{\sum_l Y_l a_l}{\sum_l Y_l} - a_j. \quad (\text{JUNCTION RETURN})$$

The resulting scattering map $S : a \mapsto b$ preserves the Y -weighted flux. With one incoming parent and daughter branches its reflected amplitude is

$$\frac{Y_{\text{parent}} - \sum_d Y_d}{Y_{\text{parent}} + \sum_d Y_d}. \quad (\text{BRANCH REFLECTION})$$

It vanishes exactly at admittance matching.

Let an edge of exact length ℓ_e carry phase $P_{e(\kappa)} = e^{i\kappa\ell_e}$ and let $P(\kappa)$ be the diagonal propagation law. A standing current is precisely a fixed current of one complete propagation and junction return:

$$SP(\kappa)a = a. \quad (\text{STANDING RETURN})$$

Along a closed path this is the phase-closure relation

$$\sum_e \kappa_e \ell_e + \sum_j \delta_j = n\Theta,$$

where Θ is one full turn and the δ_j are junction turns. In a uniform branch the least positive closure length is therefore

$$\lambda = \frac{\Theta}{|\kappa|}. \quad (\text{WAVELENGTH})$$

Finally let $W \geq 0$ be an edge conductance or stiffness and $G \geq 0$ a storage operator. For $\text{Re } z > 0$ define the dynamic incidence law

$$L(z) = d^* W d + zG. \quad (\text{DYNAMIC BODY})$$

Split its variables into interior and boundary ports and write

$$L(z) = \begin{pmatrix} A(z) & B(z) \\ C(z) & D(z) \end{pmatrix}.$$

Whenever $A(z)$ is invertible, imposing zero interior injection returns the boundary response

$$S_{B(z)} = D(z) - C(z)A(z)^{-1}B(z). \quad (\text{DYNAMIC SHORT})$$

It is positive real:

$$\text{Re} \langle S_{B(z)} y, y \rangle \geq 0 \quad \text{for } \text{Re } z > 0. \quad (\text{PASSIVE RETURN})$$

Demonstration. Pairing an additive current with the oriented boundary of one event gives (“EVENT BALANCE”). Summing those identities cancels every intermediate q_k by causal time parity and proves (“INTERVAL BALANCE”).

The Kirchhoff identity is the adjoint relation $\langle d\varphi, c \rangle = \langle \varphi, Bc \rangle$. (“CURVED LOOP”) is discrete Stokes. It also shows why a loop with changing enclosed flux has a returned interior term rather than violating conservation. At the junction, $b_j = u - a_j$. Substitution into current balance gives $u \sum_j Y_j = 2 \sum_j Y_j a_j$, proving (“JUNCTION RETURN”) and (“BRANCH REFLECTION”). In matrix form,

$$S = 21 \frac{Y}{1^* Y 1} - I,$$

so direct multiplication gives $S^* Y S = Y$. Propagation preserves the same flux when the phases have unit modulus. A standing current is therefore a unit-return eigenvector; multiplying its edge and junction phases around a closed path gives (“WAVELENGTH”).

For the dynamic body,

$$\operatorname{Re} \langle L(z)x, x \rangle = \|W^{\frac{1}{2}} dx\|^2 + \operatorname{Re}(z) \|G^{\frac{1}{2}} x\|^2 \geq 0.$$

Given boundary value y , put $x = -A(z)^{-1} B(z)y$. Then $L(z)(x \oplus y) = 0 \oplus S_{B(z)}y$, so

$$\operatorname{Re} \langle S_{B(z)}y, y \rangle = \operatorname{Re} \langle L(z)(x \oplus y), x \oplus y \rangle \geq 0.$$

This proves (“PASSIVE RETURN”).

Boundary. The oriented complex supplies legal composition and cancellation; it does not select A , W , G , Y , a metric, or a coefficient system. Those are parameters of the represented ecology. The theorem applies equally when the coefficients represent electrical charge, chemical storage, mechanical action, probability amplitude, or an analytic boundary field, provided the displayed hypotheses are actually supplied. A lossy junction gives a weighted contraction rather than a flux isometry. A globally nonexact connection may carry nontrivial loop holonomy even where its local curvature vanishes.

The theorem unifies three relations which should not be separated:

- current balance is the boundary face, including storage and exterior return;
- voltage balance is the coboundary face of receiver-relative potential; and
- wave reflection is the constitutive return at a branching boundary.

Kirchhoff’s laws therefore have a homological form before they are assigned electrical units [18]. Port-Hamiltonian graph theory makes the same incidence carrier compositional across storage, dissipation, open ports, and examples outside circuit theory [19]. The domain chooses the coefficient system and constitutive law; the elementary cancellation and gluing law remain the same.

For a metabolic body, q may carry chemical or thermal storage, j the circulating transport, and r uptake, work, waste, or environmental exchange. Homeostasis means bounded or periodic return of q over the selected order of time, not equality of every visible current at every cut. A **leak** is nonzero transport through the selected exterior boundary. A **short** is an internal low-resistance cycle which dissipates current without completing the body’s required return. Neither erases conservation in the encompassing ecology; each changes which holon closes.

Branching resource networks make the scale relation concrete. Biological allometry has long modeled material transport through space-filling fractal-like branching tubes [20]. At a pulsatile branch, the same geometry has a wave face: the parent reflects exactly to the extent that its admittance fails to equal the combined daughter admittance. A recursive branch law is therefore simultaneously a scale-rebase condition and an impedance fixed-point condition.

VIII.3 Wavelength is a return length

The full-turn symbol Θ is retained until a particular angular chart sets $\Theta = 2\pi$. No floating approximation is required. An exact branch phase can remain

$$e^{i\kappa\ell} = \sum_{n \geq 0} \frac{(i\kappa\ell)^n}{n!}.$$

A standing wave is the eigen-current which survives one complete propagation and junction return. Its wavelength is the least positive length which returns the phase:

$$e^{i\kappa\lambda} = 1, \quad \lambda = \frac{\Theta}{|\kappa|}.$$

Quantum transport is one coefficient representation of this elementary closure. Sequential events compose as operators, co-present boundaries as tensor products, alternate paths add as complex amplitudes, and the opposed causal hand is represented by the adjoint. For a complete closed ecology the return can be unitary; for an open receiver, the complete family of channels closes even though no selected channel must conserve norm by itself. Quantum graphs give a literal instance: Kirchhoff vertex conditions on a graph Laplacian produce a unitary scattering matrix assembled from its subgraphs [21]. The physical identification $p = \hbar\kappa$ supplies the action scale; it does not create the underlying phase-return geometry.

IX Metric, curvature, and changing geometry

IX.1 Metric is a local comparison law

On a smooth manifold M , a Riemannian metric g assigns a positive-definite comparison of tangent directions. Its Levi-Civita connection ∇ supplies compatible transport. Curvature measures the failure of neighboring transports to commute. These are consecutive structures, not synonyms:

Metric g	Local length, angle, volume, and tangent comparison.
Connection ∇	How a carrier is compared along a path.
Curvature R	The oriented mismatch obtained by comparing neighboring transports.
Ricci Ric	The transverse trace of directional curvature.
Ricci flow	A law by which that traced face changes the later metric.

For a geodesic family with unit tangent u and separation ξ ,

$$\nabla_u \nabla_u \xi = -R(\xi, u)u$$

under the selected sign convention. If (e_a) is an orthonormal transverse frame, then

$$\text{Ric}(u, u) = \sum_a g(R(e_a, u)u, e_a).$$

Ricci is therefore a receiver contraction of the fuller curvature operator.

IX.2 Ricci flow changes the receiver

Hamilton's equation

$$\partial_\tau g = -2 \text{ Ric}(g)$$

does not move a point through one fixed geometry. It changes the metric by which later tangent directions, volumes, and continuations are compared [22]. A curvature singularity marks failure of that smooth continuation chart. Rescaling a neighborhood can expose a recurrent local model; a surgery law, when its hypotheses hold, is a separate discrete deed.

Perelman's work supplies the needed monotone quantities, non-collapsing control, singularity analysis, and surgery theory [23], [24]. The Poincare theorem then provides a rigorous local-to-global example: a closed simply connected smooth 3-manifold is diffeomorphic to S^3 , with the topology recovered through the controlled flow and surgery account [25].

The elementary holonic lesson is exact but modest: local metric evolution, singular rebase, component departure, and retained global topology can form one proof. Smooth flow, surgery, topological identity, and metric appearance remain different relations.

IX.3 Lightning as a path which changes later transport

The lightning comparison belongs here because the leader does not merely traverse a passive coordinate space. Ionization, charge redistribution, channel heating, conductivity, and geometry change the medium seen by the next leader and by the return stroke. Leader-return-stroke models explicitly couple the preceding channel state to the later current and radiation [26].

In holonic language:

field + medium $\rightarrow$ leader formation $\rightarrow$ changed conductive terrain $\rightarrow$ attachment $\rightarrow$ return current $\rightarrow$ later changed field.

The correspondence is about path-written transport geometry. A literal physical model still uses its actual electromagnetic, plasma, chemical, thermal, and fluid laws. Holonics supplies the incidence and transport grammar by which those laws can be related without replacing them.

X Duality, dimension, and the open M

X.1 A duality is a partial transition

Let two local descriptions be M_α and M_β . A situated duality is a supplied map

$$D_{\alpha\beta} : U_{\alpha\beta} \subset M_\alpha \rightarrow U_{\beta\alpha} \subset M_\beta$$

for which named consequential diagrams commute on the actual overlap. Its domain, approximation, orientation, and receiver family are part of the relation.

The network of such descriptions has no completed external chart. Any proposed total view is another situated receiver. In the project's language, M names this open, growing atlas of local descriptions and transitions, not an absolute container holding every perspective at once.

X.2 Worldline, worldsheet, and worldvolume

A lineage is string-like because it is an ordered one-parameter arc. A co-present transverse family is sheet-like because it retains relations among several arcs without inventing a serialization. Transporting the sheet through another parameter sweeps a volume-like construction. These are dimensions of incidence and record before they are claims about physical strings.

String theory is relevant because its duality web gives exact physical cases in which descriptions with apparently different elementary objects, dimensions, radii, couplings, momentum, and winding can represent the same selected consequences on a stated overlap. Witten's relation between strongly coupled Type IIA theory and an eleven-dimensional supergravity limit is one central example [27].

The sigma-model bridge is even more direct. At leading order, the metric beta function contains the Ricci tensor, so target geometry changes with renormalization scale [28]. T-duality is a separate relation transporting a background to another chart. Flow and duality can commute under specific hypotheses, but neither is a universal replacement for the other.

Holonics extracts the mathematical discipline exhibited by these theories:

- a local description is not absolute;
- the transition has a domain;
- transported observables determine what is shared;
- winding and dimension can change their presented roles; and

- a loop through descriptions can retain nontrivial return.

XI Compression, probability, and loss

XI.1 Compression is factorization through future questions

Let $q : X \rightarrow Q$ be an emitted face and let $\mathcal{O}$ be a declared family of future observations. The face is receiver-exact precisely when every $o \in \mathcal{O}$ factors:

$$\exists |o| : Q \rightarrow Y_o \quad \text{such that} \quad o = |o| \circ q.$$

Compression is therefore not fundamentally byte count or geometric volume. It is the factorization of distinctions which are inactive for the declared future relation. When a later receiver can distinguish two members of one fiber of q , the earlier compression was exact only for the smaller family.

Every event presented through a receiver emits a face. That face is a nontrivial compression only where q identifies distinctions while the declared future observations still factor through it. An identity receiver is therefore a lawful uncompressed limiting case. A phase transition occurs when the current factorization law ceases to continue and a new discrete invariant or cell type must be carried.

XI.2 Probability is a first-person quotient

Let an observer carry a filtration $(\mathcal{F}_t)$ of what has become available on a probability space $(\Omega, \mathcal{F}, \mu)$. For an event E , the receiver-relative conditional field is the $\mathcal{F}_t$ -measurable random variable

$$\Pr_\rho(E \mid \mathcal{F}_t) = E_\mu[\mathbf{1}_E \mid \mathcal{F}_t],$$

characterized by

$$\int_A \Pr_\rho(E \mid \mathcal{F}_t) d\mu = \mu(E \cap A) \quad \text{for every} \quad A \in \mathcal{F}_t.$$

If the contemporary receiver is further restricted to one admitted event $A_t \in \mathcal{F}_t$ with $\mu(A_t) > 0$, the scalar face is the quotient

$$\Pr_\rho(E \mid A_t) = \frac{\mu(E \cap A_t)}{\mu(A_t)}.$$

Bayes' theorem rebases this scalar quotient after a new event changes the conditioning region:

$$\Pr(H \mid E) = \frac{\Pr(E \mid H) \Pr(H)}{\Pr(E)}.$$

The numbers do not cause the paths and do not declare truth. They summarize recurrence and admitted possibility from the observer's contemporary position. A deterministic ecology can produce the same conditional field under the same circumstances while remaining probabilistic from the first-person receiver that does not contain the complete exterior.

XI.3 Loss is the transported residual

For a comparison square with two routes γ_1, γ_2 , the primary loss is the failure to commute:

$$\mathcal{L}_\rho = X(\gamma_1, \gamma_2) = (T_{\gamma_1}, T_{\gamma_2}).$$

If an additive chart is supplied, it may be represented as a difference. If a norm is supplied, that difference may be projected to a nonnegative scalar. Those are later receiver faces.

Loss is consequently information about changed relation. It is neither reward nor punishment and does not imply that one route exists solely to be corrected. An OPEN residual can become terrain for a later construction.

XII The elements form a network

The reusable library introduced with this paper is organized by mathematical role rather than by application:

Definitions	Situated occurrence; receiver; holon; comparison face; phase; situated event; holonic process double category; causal soul; situated formulation family; formulation-span atlas; causal time parity.
Lemmas	Ordered composition; receiver non-reconstruction; pullback composition of formulation spans.
Theorems	Comparison-boundary closure; transported fundamental theorem; causal-parity interval balance, Kirchhoff duality, and phase return; soul-sameness hierarchy; positive perspective descent; natural contraction feasibility set; exact π split; exact e split; receiver-qualified one-prime Euler formulation cell.
Corollaries	Point-line-loop as receiver-relative faces; complete natural zeta contraction implies RH.

Each object carries a stable key, dependency list, status, claim, proof, and boundary. A paper imports the object and places it inside a larger argument. The paper may add a geometric demonstration or an application-specific specialization; it does not become the definition's owner.

XII.1 Immediate extensions

The first formulation layer is now exact: receiver-exact correspondences compose, while noninvertible quotients remain outside the rechart core. Further objects should be added only when their exact statements are ready:

1. a theorem for receiver-exact compression under a receiver family which itself changes by lawful base transport;
2. a nonlinear or sheaf-valued extension of the causal Stokes and port-return theorem;
3. a local connection and curvature object that supports nonlinear partial transports;
4. a retriangulation theorem separating combinatorial, projective, conformal, and metric invariants;
5. a holomorphic return object carrying conjugacy, multiplier, critical orbit, and residual;
6. a Bayesian rebase object over an explicitly supplied filtration and measure; and
7. a phase-boundary object joining continuous deformation to discrete change of cell type.

These are mathematical continuations. None requires machine-learning terminology.

XII.2 Applications return as specializations

Once the elementary objects are stable, an application chooses its own medium and laws:

- number theory supplies divisibility, valuations, analytic continuation, and zeta;
- physics supplies metrics, connections, constitutive laws, units, and conservation;
- language supplies inscription, speech, grammar, reference, and consequence;
- computation supplies algorithms, state transitions, storage, and physical execution; and
- machine learning supplies parameterized transformations, conditioning, routing, and update laws.

Each can be represented as relational geometry because each has occurrences, boundaries, transport, comparison, and return. Their domain laws remain indispensable. Holonics does not replace the sciences; it exposes the common elementary architecture by which their constructions can be related.

XIII Conclusion

The foundational object is not intelligence, a model, a number, a particle, or a file. It is a situated relation in motion.

An occurrence participates in incidence. Transport carries it through an ordered path. A receiver presents one face. A comparison cell relates alternate paths. A holon closes relative to an exposed boundary and can become a constituent at another grain. An atlas grows through actual overlaps. Holomorphy supplies a precise scale-turn chart. Fractal recurrence supplies return and rebase. Differential geometry supplies local comparison, curvature, and changing metric. Duality supplies lawful re-expression without an external master view. Probability is a receiver quotient of admitted recurrence; loss is a receiver face of transported residual; compression is the successor's contextual factorization.

This is the intended meaning of emergent complexity as a scientific framework. Rich global structure does not need a different elementary ontology for every application. It needs exact local relations, carried differences, and a mathematics capable of composing them without flattening their interiors.

Bibliography

- [1] O. Byrne, *The First Six Books of the Elements of Euclid, in Which Coloured Diagrams and Symbols Are Used Instead of Letters for the Greater Ease of Learners*. London: William Pickering, 1847. [Online]. Available: <https://archive.org/details/firstsixbooksof00byrn>
- [2] S. Slyusarev, "Byrne: A LaTeX and MetaPost Package for Geometric Proofs." Accessed: Jul. 23, 2026. [Online]. Available: <https://github.com/jemmybutton/byrne-latex>
- [3] A. Hatcher, *Algebraic Topology*. Cambridge University Press, 2002. [Online]. Available: <https://pi.math.cornell.edu/~hatcher/AT/AT.pdf>
- [4] U. Pachner, "P.L. Homeomorphic Manifolds are Equivalent by Elementary Shellings," *European Journal of Combinatorics*, vol. 12, no. 2, pp. 129–145, 1991, doi: [10.1016/S0195-6698\(13\)80080-7](https://doi.org/10.1016/S0195-6698(13)80080-7).
- [5] E. W. Weisstein, "Regular Hexagon." Accessed: Jul. 25, 2026. [Online]. Available: <https://mathworld.wolfram.com/RegularHexagon.html>
- [6] E. W. Weisstein, "Pentagonal Pyramid." Accessed: Jul. 25, 2026. [Online]. Available: <https://mathworld.wolfram.com/PentagonalPyramid.html>
- [7] E. W. Weisstein, "Great Dodecahedron." Accessed: Jul. 25, 2026. [Online]. Available: <https://mathworld.wolfram.com/GreatDodecahedron.html>
- [8] E. W. Weisstein, "Small Stellated Dodecahedron." Accessed: Jul. 25, 2026. [Online]. Available: <https://mathworld.wolfram.com/SmallStellatedDodecahedron.html>
- [9] J. Garrido-García and others, "A Natural Programmable Metamaterial Controls 3D Curvature of Compound Eyes," *Nature Communications*, vol. 17, p. 5553, 2026, doi: [10.1038/s41467-026-74276-6](https://doi.org/10.1038/s41467-026-74276-6).
- [10] A. Pande and others, "Hunting Ecology Predicts Eye Arrangements in the Modular Visual System of Spiders," *Current Biology*, 2026, doi: [10.1016/j.cub.2026.06.019](https://doi.org/10.1016/j.cub.2026.06.019).
- [11] E. M. Jakob *et al.*, "Lateral Eyes Direct Principal Eyes as Jumping Spiders Track Objects," *Current Biology*, vol. 28, no. 18, pp. R1092–R1093, 2018, doi: [10.1016/j.cub.2018.07.065](https://doi.org/10.1016/j.cub.2018.07.065).

- [12] OpenAI, “Planar Point Sets with Many Unit Distances.” Accessed: Jul. 25, 2026. [Online]. Available: <https://cdn.openai.com/pdf/74c24085-19b0-4534-9c90-465b8e29ad73/unit-distance-proof.pdf>
- [13] J. Milnor, *Dynamics in One Complex Variable*, Third. Princeton University Press, 2006.
- [14] T. Lei, “Similarity Between the Mandelbrot Set and Julia Sets,” *Communications in Mathematical Physics*, vol. 134, pp. 587–617, 1990, doi: [10.1007/BF02098448](https://doi.org/10.1007/BF02098448).
- [15] C. T. McMullen, “The Mandelbrot Set Is Universal,” *The Mandelbrot Set, Theme and Variations*, pp. 1–17, 2000, [Online]. Available: <https://people.math.harvard.edu/~ctm/papers/home/text/papers/muniv/muniv.pdf>
- [16] P. Doyle and C. McMullen, “Solving the Quintic by Iteration,” *Acta Mathematica*, vol. 163, pp. 151–180, 1989, doi: [10.1007/BF02392735](https://doi.org/10.1007/BF02392735).
- [17] NIST Digital Library of Mathematical Functions, “Exponential and Logarithmic Functions.” Accessed: Jul. 23, 2026. [Online]. Available: <https://dlmf.nist.gov/4>
- [18] D. H. Delphenich, “Homology, Equilibrium, and Conservation Laws I: Discrete Systems of Points,” 2019, [Online]. Available: <https://arxiv.org/abs/1901.03171>
- [19] A. J. van der Schaft and B. M. Maschke, “Port-Hamiltonian Systems on Graphs,” *SIAM Journal on Control and Optimization*, vol. 51, no. 2, pp. 906–937, 2013, doi: [10.1137/110840091](https://doi.org/10.1137/110840091).
- [20] G. B. West, J. H. Brown, and B. J. Enquist, “A General Model for the Origin of Allometric Scaling Laws in Biology,” *Science*, vol. 276, no. 5309, pp. 122–126, 1997, doi: [10.1126/science.276.5309.122](https://doi.org/10.1126/science.276.5309.122).
- [21] V. Kostykin and R. Schrader, “Kirchhoff’s Rule for Quantum Wires,” *Journal of Physics A: Mathematical and General*, vol. 32, no. 4, pp. 595–630, 1999, doi: [10.1088/0305-4470/32/4/006](https://doi.org/10.1088/0305-4470/32/4/006).
- [22] R. S. Hamilton, “Three-Manifolds with Positive Ricci Curvature,” *Journal of Differential Geometry*, vol. 17, no. 2, pp. 255–306, 1982, doi: [10.4310/jdg/1214436922](https://doi.org/10.4310/jdg/1214436922).
- [23] G. Perelman, “The Entropy Formula for the Ricci Flow and Its Geometric Applications.” [Online]. Available: <https://arxiv.org/abs/math/0211159>
- [24] G. Perelman, “Ricci Flow with Surgery on Three-Manifolds.” [Online]. Available: <https://arxiv.org/abs/math/0303109>
- [25] J. Morgan and G. Tian, *Ricci Flow and the Poincare Conjecture*. American Mathematical Society, Clay Mathematics Institute, 2007. [Online]. Available: <https://www.claymath.org/resource/ricci-flow-and-the-poincare-conjecture/>
- [26] S. Cai, V. A. Rakov, N. Liu, and others, “A Leader-Return-Stroke Consistent Macroscopic Model for Calculations of Return Stroke Current and Its Optical and Electromagnetic Emissions,” *Journal of Geophysical Research: Atmospheres*, vol. 122, pp. 8686–8704, 2017, doi: [10.1002/2017JD026490](https://doi.org/10.1002/2017JD026490).
- [27] E. Witten, “String Theory Dynamics in Various Dimensions,” *Nuclear Physics B*, vol. 443, no. 1–2, pp. 85–126, 1995, doi: [10.1016/0550-3213\(95\)00158-O](https://doi.org/10.1016/0550-3213(95)00158-O).
- [28] D. H. Friedan, “Nonlinear Models in Two Plus Epsilon Dimensions,” *Annals of Physics*, vol. 163, no. 2, pp. 318–419, 1985, doi: [10.1016/0003-4916\(85\)90384-7](https://doi.org/10.1016/0003-4916(85)90384-7).